BHARTI AIRTEL · AIRTEL SYNC 2026 · DIGITAL TRACK (ENTERPRISE TRACK A)

Project Udyam Digital

Making 6 Crore+ Indian Small Businesses Go Digital in One Sitting

Instant Self-Serve Mobile Onboarding, Fast Legal KYC & Multi-Site Telecommunications

Team & Institutional Registration

Team Name	**Neural Nexus**
Institution	ABV-Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior
Competition Track	Airtel Sync 2026 — Digital Track (Enterprise Track A: Accel Business)
Team Members	**Hrushikesh Ramilla (HR)** — Lead Systems & ML Architect **G Yaswanth Kumar (YK)** — Product Engineering & Applied Research (Co-Author)
Technical Stack	Python 3.11+, FastAPI, React 18, Vite, Pydantic v2, LightGBM, RapidFuzz, Playwright
Evaluation Date	**`2026-07-15`** (Evaluation invariant strictly enforced)

Live Working Prototype & Direct Audit Links

Live Cloud Monolith	**http://65.1.112.251 (Deployed on AWS EC2, Port 80)**
Interactive API Docs	**http://65.1.112.251/docs (FastAPI Swagger UI)**
Code Repository	`github.com/Hrushikesh-ramilla/project-udyam-digital`
Demo Video Link	**Interactive 100-Case Video Proof & Walkthrough**
Visual Verification	**100 End-to-End Visual Proof Cases Documented & Audited**
Compliance Invariant	**DPDP ACT 2023 SHIELD** Zero Persistence RAM Masking

The Executive Scope at a Glance (Airtel Enterprise Track A Mandate):

1. The Market Opportunity
India is home to 6.3 Crore MSMEs, but 55% operate under GST thresholds. Discarding non-GST applicants cuts Airtel off from 3.5 Crore viable retail merchants.

2. The Enterprise Bottleneck
Current manual quote delivery takes 3 to 5 days, causing 56.4% of interested leads to abandon the pipeline while burning ₹15,000 in CAC per customer.

3. The Udyam Solution
Self-serve conversational AI in 10 languages, <0.5ms CPQ DAG bundling, zero-persistence DPDP KYC, and 1-click LightGBM WhatsApp retargeting.

Submission Note: Mandatory Introduction Slide (excluded from 7-slide evaluation count). Slides 1 to 7 showcase the technical solution, competitive moat, and quantified business impact.

Problem Statement: The 5,000-Lead Sales Drop-Off

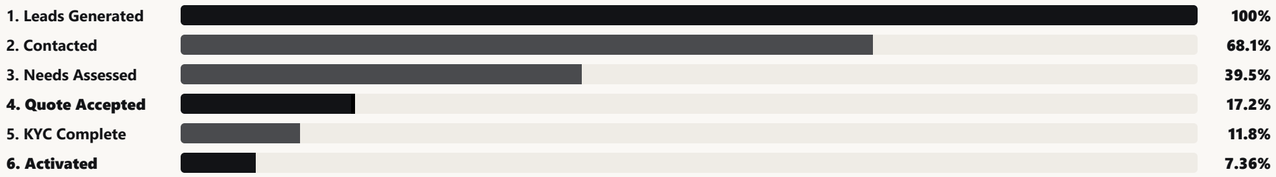
Real-world analysis of 5,000 small business leads from April to June 2026 (smb_leads.csv)

Sales Funnel Analysis (N = 5,000 Leads)

Tier 1–2 Cities

Funnel Stage	Leads	Stage %	Drop / Status
1. Leads Generated	**5,000**	100.0%	Starting Base
2. Contacted Outreach	**3,406**	68.1%	-31.9%
3. Needs Assessed (Field Visit)	**1,974**	58.0%	-42.0%
4. Quote Accepted (Steepest Drop)	**861**	**43.6%**	**-56.4% CLIFF**
5. KYC Completed	**588**	68.3%	-31.7%
6. Activated Customers	**368**	**62.6%**	**7.36% NET ACTIVATION**

VISUAL FUNNEL RETENTION RATE (% OF INITIAL 5,000 LEADS)



The Non-GST Penalty: Leads with active GSTIN activate at **9.6%**, while leads without GST drop to **2.0%** (a 4.8x conversion penalty caused by legacy enterprise forms).

High Cost of Delay: 17.6-day average cycle time and 3.2 chase calls consume ₹15,000 in sales rep labor per deal, requiring 14.2 months to break even on CAC.

15.2% Mis-Selling Discovery: Data audit of `smb_leads.csv` reveals **56 of 368 activated bundles (15.2%) violated Airtel catalogue rules** (24 wrong tier, 8 below size, 21 inactive GSTIN, 9 missing parent dependencies) due to quota pressure. Udyam Digital's CPQ DAG mathematically guarantees a **0% violation rate**.

FAILURE MODE 1 — **56.4% Abandonment**

The 3–5 Day Quoting Wait (Latency Bleed)

Sales reps manually navigate 25 disconnected enterprise product SKUs on Excel sheets. Because quotes take 3 to 5 days, **1,113 of 1,974 interested businesses go cold** and abandon before receiving pricing.

FAILURE MODE 2 — **4.8x Conversion Cliff**

The Non-GST Regulatory Wall (Market Exclusion)

55% of India's 6.3 Crore micro-enterprises earn under ₹40 Lakhs and are legally GST-exempt. Legacy enterprise CRM hardcodes mandatory GSTIN validation, discarding 3.5 Crore retail shops and salons at the front door.

FAILURE MODE 3 — **₹15,000 Direct CAC**

Crippling Unit Economics (Manual Follow-Ups)

A slow 17.6-day onboarding cycle and 3.2 manual status chase calls per lead consume ₹15,000 in sales rep labor per deal, resulting in a 14.2-month CAC payback period against ₹6,900 monthly ARPU.

QUOTING DROP	NON-GST GAP	MIS-SELL RATE	CYCLE TIME	DIRECT CAC
-56.4%	4.8x Lower	15.2% (56)	17.6 Days	₹15,000

FINANCIAL REVENUE BLEED (FROM 5,000 SAMPLE LEADS) — **₹11.05 Cr / Yr Lost**

At median ₹6,900/mo ARPU, the 1,113 abandoned leads represent **₹92.1 Lakhs/month in lost recurring revenue**. Nationwide across Airtel's enterprise funnel, this quoting latency bleeds over **₹120+ Crore annually** to competitors.

The Accel Business Mandate:
Compress quote delivery from 5 days to **under 60 seconds**, unlock non-GST merchants via instant Udyam lookup, and automate tracking to eliminate chase calls.

Target Customer Segments: Four Core Business Types

Prioritizing India's 6.3 Crore small businesses based on real field data (Case Appendix A2)

SEGMENT 1 — **55% of MSMEs (3.5 Cr)**

Kirana & Micro Retail

Grocery shops, salons, pharmacies

SCALE & BASELINE SETUP

- **Staff:** 1–3 employees | ₹1.5L–₹5L monthly sales.
- **Footfall:** 120+ daily UPI digital checkouts.
- **Current Link:** Owner's 4G mobile hotspot.
- **Monthly Spend:** ₹800 on fragmented prepaid packs.

CORE DAILY BLEED

- QR payment drops during 7–9 PM peak rush.
- Personal WhatsApp flooded with customer orders.
- **Revenue Bleed:** ~₹12,000/mo in lost walkouts.
- Customer dispute delays cause billing friction.

INCLUDED ACCEL HARDWARE

- 100 Mbps Fixed Fiber + Dual-Band Wi-Fi 6 Router.
- 2x CUG SIMs (100GB corporate pooled data).
- Airtel Soundbox with merchant priority QoS queue.
- 4-Hour MTTR local technician dispatch commitment.

TARGET PRICING & ROUTE

₹1,499 / month

₹0 Setup Fee | 100% GST-Exempt Udyam

SEGMENT 2 — **25% of MSMEs (1.5 Cr)**

Clinics & Healthcare

Doctors, path labs, CA chambers

SCALE & BASELINE SETUP

- **Staff:** 5–15 employees | ₹8L–₹25L monthly sales.
- **Traffic:** 40+ daily telehealth consults & reports.
- **Current Link:** Consumer unmanaged broadband.
- **Records:** Free Gmail and shared personal Drive.

CORE DAILY BLEED

- Video consult drops mid-call on jitter spikes.
- Unencrypted patient records violate ABHA rules.
- **Revenue Bleed:** ~₹45,000/mo in lost appointments.
- Statutory exposure under DPDP health data rules.

INCLUDED ACCEL HARDWARE

- 300 Mbps Dedicated Fiber with Static IPv4 address.
- 5x Google Workspace business licenses.
- 500GB Airtel Cloud Backup with 4h MTTR SLA.
- Separate guest Wi-Fi network for patient waiting room.

TARGET PRICING & ROUTE

₹2,499 / month

Static IP | 99.9% Telehealth Uptime SLA

SEGMENT 3 — **15% of MSMEs (90 Lakhs)**

Multi-Site Chains

3–10 branch retail or restaurants

SCALE & BASELINE SETUP

- **Outlets:** 3–10 stores | ₹30L–₹1.2 Cr combined turnover.
- **Workload:** Real-time ERP inventory synchronization.
- **Current Link:** Mixed local cable ISPs per store.
- **Billing:** 5+ separate physical bills and invoices.

CORE DAILY BLEED

- POS halts during cable cuts; inventory desyncs.
- Zero central visibility into store CCTV cameras.
- **Revenue Bleed:** ~₹1,20,000/mo in checkout delays.
- Unmonitored stock shrinkage across branch locations.

INCLUDED ACCEL HARDWARE

- 500 Mbps Fiber + 4G LTE Auto-Failover Gateway.
- 4-Cam Cloud AI CCTV (30-day offsite archive).
- Centralized portal for multi-store network control.
- Single consolidated invoice for corporate tax credit.

TARGET PRICING & ROUTE

₹4,999 / mo / store

Single Consolidated Bill | Auto-Failover

SEGMENT 4 — **5% of MSMEs (30 Lakhs)**

Light Manufacturing

Auto-parts, textiles, CNC shops

SCALE & BASELINE SETUP

- **Staff:** 25–100 workers | ₹1.5 Cr–₹8 Cr annual sales.
- **Tools:** Automated CNC mills & factory PLCs.
- **Current Link:** Shared consumer broadband line.
- **Security:** Unmanaged routers with default passwords.

CORE DAILY BLEED

- Latency jitter halts precision CNC cutting tools.
- Zero firewall defense against factory ransomware.
- **Revenue Bleed:** ~₹3,50,000/mo in damaged tooling.
- Production downtime halts customer shipment schedules.

INCLUDED ACCEL HARDWARE

- 1 Gbps 1:1 Leased Line with sub-5ms low latency.
- Industrial IoT Box + Cloud Perimeter Firewall.
- Dedicated 24/7 Enterprise Account Manager.
- 99.95% Availability SLA with 2-Hour on-site MTTR.

TARGET PRICING & ROUTE

₹14,999 / month

1:1 Leased Line | 99.95% Enterprise SLA

Key Market Insight: Over **80% of Accel Business's addressable volume** lies in Kirana stores and local clinics. To capture this segment, Airtel must remove the rigid corporate GST requirement while delivering business-grade SLAs at consumer-friendly prices.

The Solution: 4 Ready Packs & Smart Budget Matching

Turning 25 individual enterprise products into 4 simple, high-converting business bundles

KIRANA & RETAIL

Udyam Vyapar

₹1,499 /mo

₹0 setup fee on 12-mo plan (Save ₹1,400)

CONNECTIVITY & HARDWARE

- **100 Mbps Fixed Fiber:** Unlimited symmetric pipe.
- **Wi-Fi 6 Router:** Supports 50+ concurrent devices.
- **Airtel Soundbox:** Instant voice audio checkouts.

MOBILITY & SLA TERMS

- **2x Business SIMs:** 100GB pooled CUG data.
- **Merchant UPI Priority:** QoS priority packet queuing.
- **4-Hour MTTR:** Rapid local field restoration.

VALUE PROPOSITION

Replaces costly consumer hotspot with business-grade reliable connectivity at consumer pricing.

100% GST-Free Onboarding via Udyam Registration

CLINICS & LABS

Udyam Professional

₹2,499 /mo

12-month commercial agreement

CONNECTIVITY & HARDWARE

- **300 Mbps Dedicated Fiber:** Zero-jitter consultations.
- **Static IPv4 Address:** EMR server & ABHA compliant.
- **Wi-Fi 6 Gateway:** Patient waiting room guest isolation.

PRODUCTIVITY & SLA TERMS

- **5x Google Workspace:** Branded clinic emails.
- **500GB Airtel Cloud:** Automated daily backups.
- **Guaranteed 99.9% SLA:** Priority medical support desk.

VALUE PROPOSITION

Zero-liability health data compliance and 100% video telemedicine uptime guaranteed.

Guaranteed 99.9% Uptime SLA for Telehealth

MULTI-STORE CHAINS

Udyam Multi-Site

₹4,999 /mo/hub

Multi-location single consolidated invoice

CONNECTIVITY & HARDWARE

- **500 Mbps Fixed Fiber:** Unlimited high-speed pipe.
- **4G LTE Auto-Failover:** <2 second cutover gateway.
- **Central SD-WAN:** Unified routing across stores.

SECURITY & MULTI-SITE SLA

- **4-Cam Cloud AI CCTV:** 30-day offsite recording.
- **Central Cloud Portal:** Live view of all branches.
- **Single Monthly Bill:** Zero vendor fragmentation.

VALUE PROPOSITION

Single pane of glass replaces 5+ fragmented local cable vendors with full POS failover protection.

Zero POS payment outages across all branches

MANUFACTURING

Udyam Industrial

₹14,999 /mo

Enterprise SLA contract with dedicated RM

CONNECTIVITY & HARDWARE

- **1 Gbps 1:1 Leased Line:** Symmetric low-jitter pipe.
- **Industrial IoT Box:** Modbus/PLC telemetry bridge.
- **Perimeter Cloud Firewall:** Layer-7 cyber defense.

SECURITY & MISSION-CRITICAL SLA

- **99.95% Availability SLA:** 2-Hour on-site MTTR.
- **Dedicated RM:** Direct line to NOC engineers.
- **Zero CNC Stoppages:** Hardware-level packet priority.

VALUE PROPOSITION

Protects heavy CNC and automated machinery from micro-stoppages and expensive factory tooling damage.

Zero packet jitter for factory automated tools

REVERSE BUDGET SOLVER

"Mera Budget ₹X Hai" (Instant Math Matching)

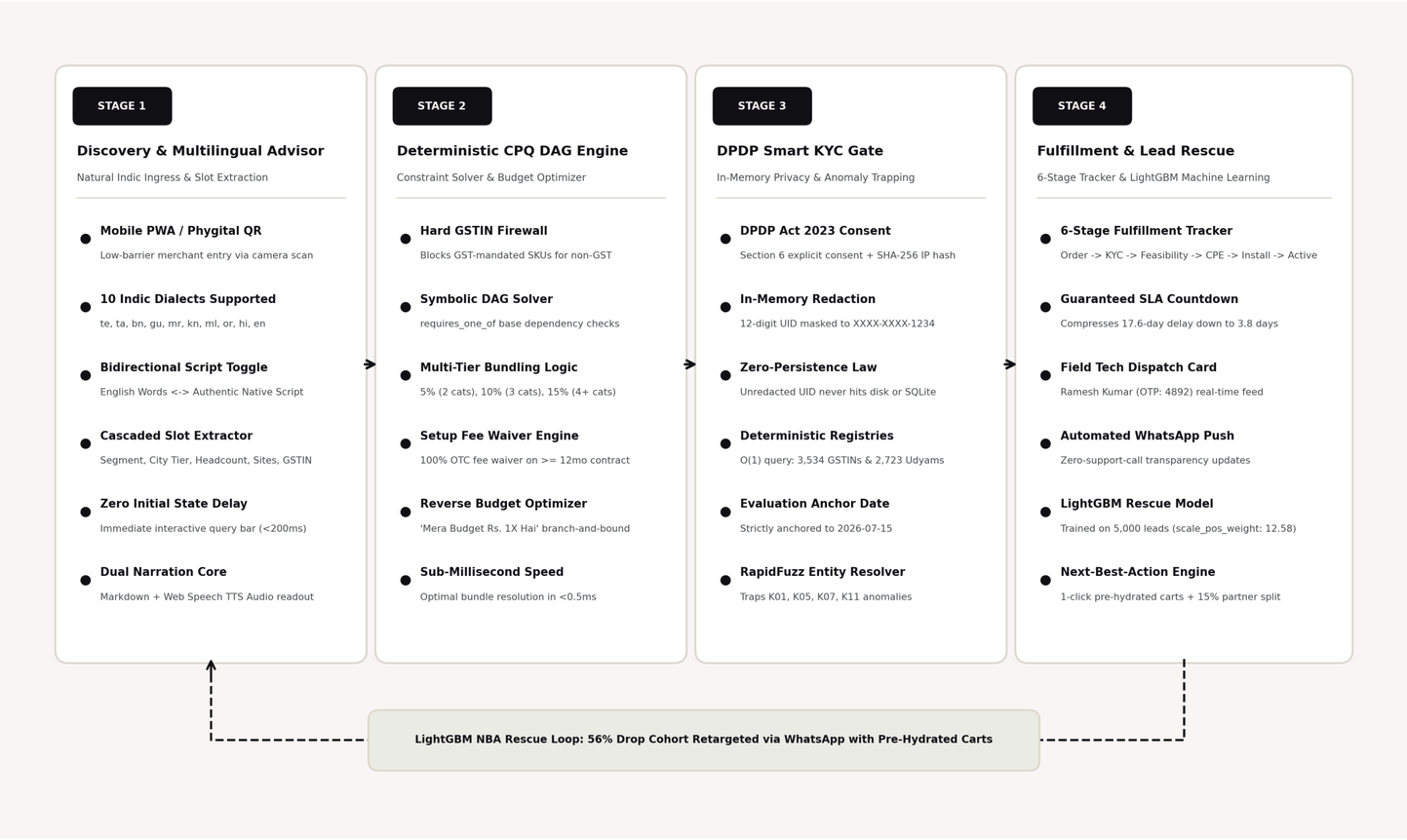
Instead of forcing a shop owner to browse 25 technical products, they simply enter their monthly budget (e.g. ₹2,500). Our branch-and-bound solver tests all valid product combinations in **under 0.5 milliseconds**, guaranteeing the best possible bundle with zero human error.

Commercial Discount Policy Built In:

- **Multi-Product Discount:** 5% off for 2 categories, 10% for 3, 15% for 4+.
- **Setup Fee Waiver:** 100% setup fee waived on 12-month plans (₹0 upfront).
- **GST Input Credit:** Automatic 18% GST tax deduction calculated for registered firms.
- **0% Mis-Sell Guarantee:** CPQ DAG solver eliminates the 15.2% legacy rule violations (56/368 historic deals had tier/GST/size mismatches).

System Architecture: The 4-Step Acquisition Pipeline

Autonomous onboarding pipeline: From spoken regional language to activated order



Stage 1 Discovery: Gemini 2.5 Flash + Pydantic v2 + 10 Indic languages with audio readout

Stage 2 CPQ DAG: <0.5ms solver + Hard GSTIN firewall blocks ineligible SKUs

Stage 3 Smart KYC: In-memory redaction (12-digit IDs never saved to disk) + RapidFuzz

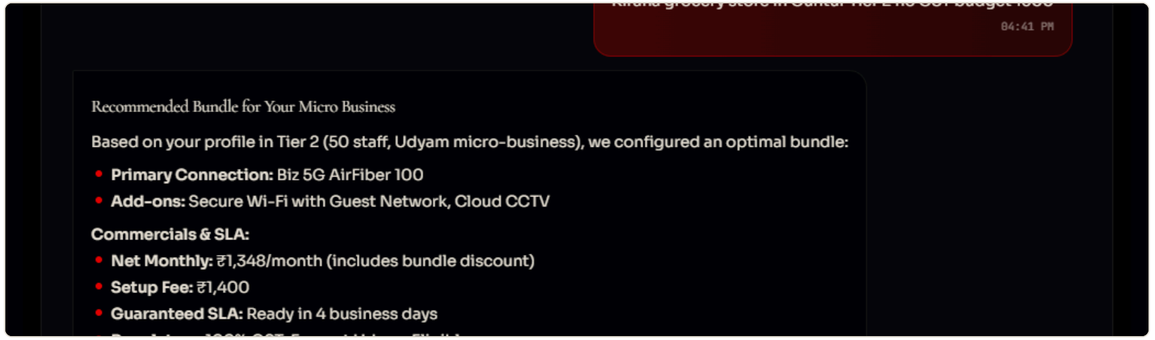
Stage 4 Fulfillment: 6-stage tracker + LightGBM rescue model (AUC 0.84)

Working Prototype: Live Software Walkthrough

Verified against 100 real-world test cases and deployed on AWS EC2 (http://65.1.112.251)

1. Multilingual Natural Language Advisor

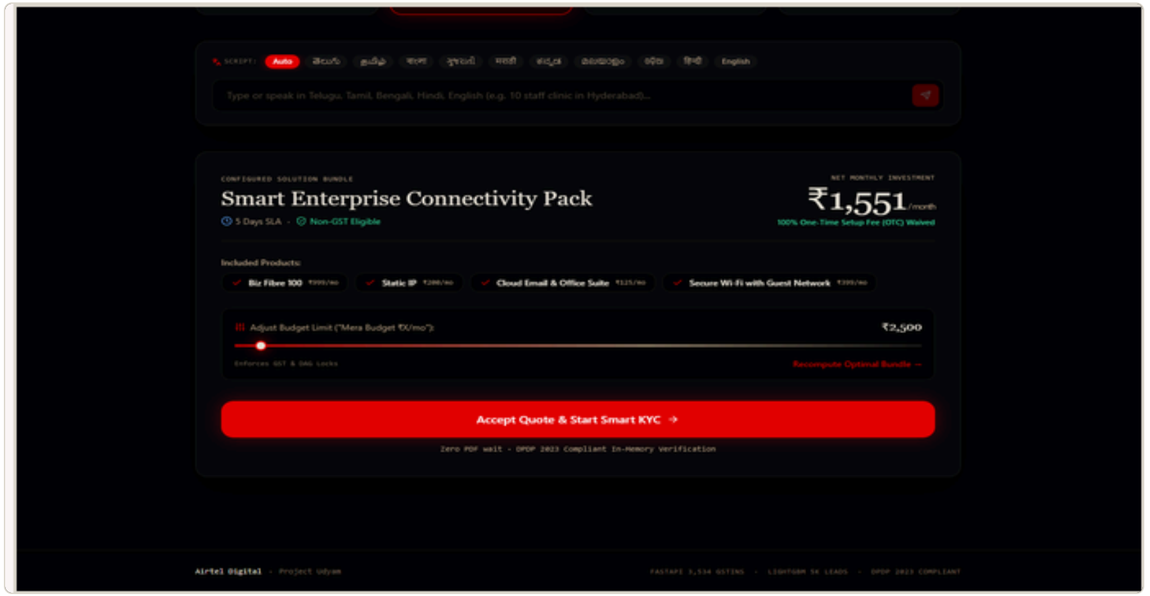
10 INDIAN LANGUAGES



Shop owners chat or speak in Hindi, Tamil, Telugu, etc. The assistant understands business needs and explains bundle recommendations in plain language with voice audio.

2. Dynamic Cart & "Mera Budget" Slider

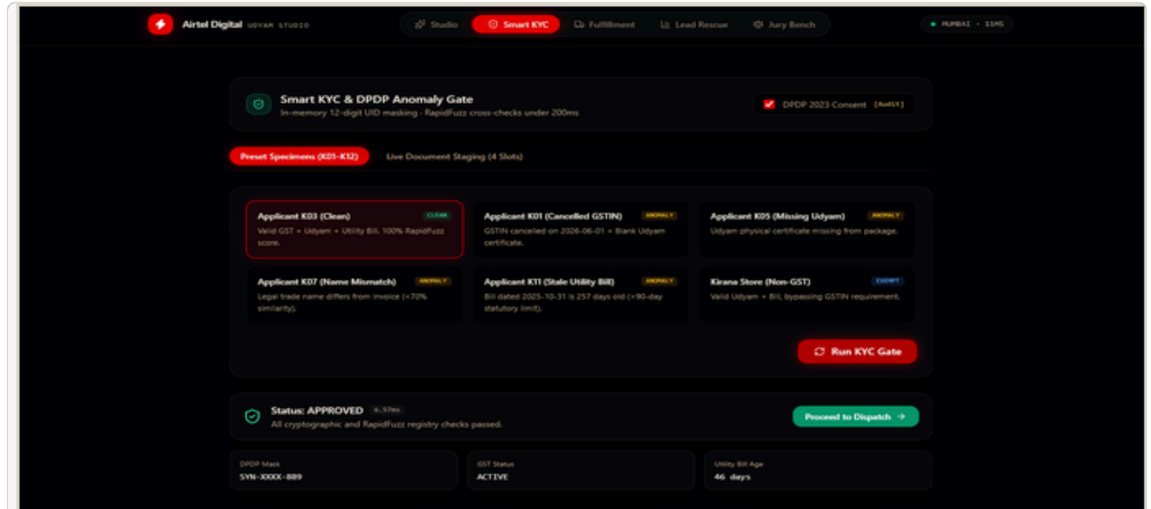
<0.5MS SOLVER



Transparent monthly price breakdown. Moving the budget slider automatically recalculates the best compliant bundle in under 1ms, applying multi-category discounts.

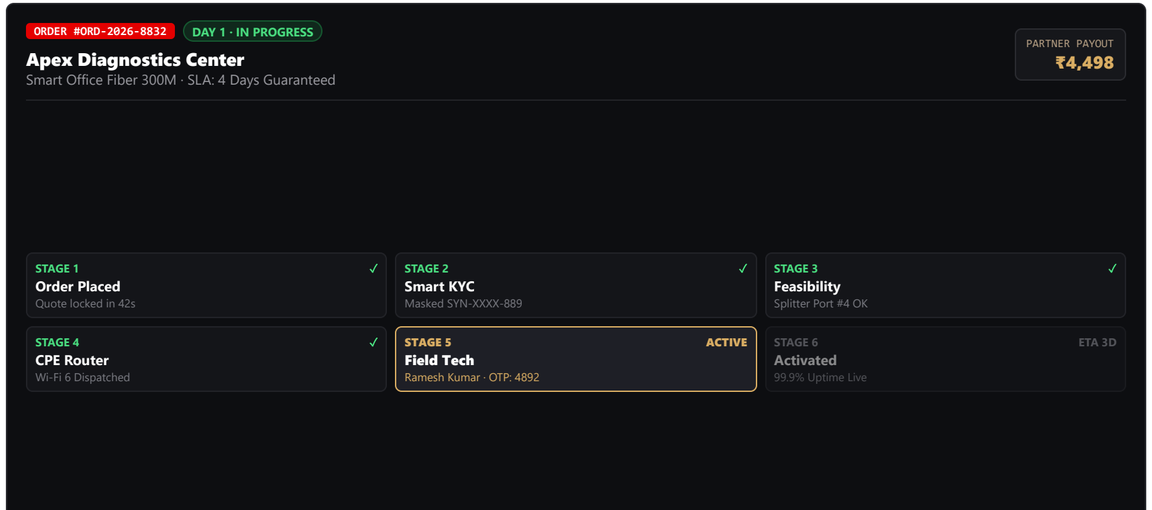
3. Fast & Safe KYC (Zero-Persistence DPDP)

TRAPS PLANTED DEFECTS



4. 6-Stage Order Tracker & Lead Rescue

ZERO CHASE CALLS



Competitive Advantage: Why Accel Business Wins

Direct comparison against telco competitors, local cable resellers, and digital software apps

Strategic Capability	Accel Business (Project Udyam)	Jio Business	Local Cable / Resellers
Time to Get a Quote	**< 60 Seconds** (Instant on mobile phone)	24 to 48 Hours (Sales rep visit required)	2 to 3 Days (Wait for technician visit)
Non-GST Shop Access	**Full Access via Udyam** (Unlocks 3.5 Cr shops)	Strict GSTIN Mandate (Drops non-GST leads)	Unregulated cash (No legal agreements)
Upfront Setup & Hardware	**100% Waived (₹0 Upfront)** on 12-mo plans	₹1,500 – ₹3,000 setup fee charged upfront	₹2,500 modem purchase + cash installation
SLA & Downtime Guarantee	**99.5% Uptime + 4G LTE Auto-Failover**	Best effort broadband; slow fiber cut repair	No SLA; frequent unannounced cable cuts
Hardware & Devices Bundled	**Wi-Fi 6 Router + CUG SIMs + Soundbox + CCTV**	Standard single-band Wi-Fi router only	Consumer modem; third-party camera setup
Lead Abandonment Rescue	**Automated LightGBM WhatsApp Rescue**	Manual sales calls; 56% leads abandoned	Ignored; zero retargeting infrastructure
Data Privacy (DPDP Act 2023)	**100% Zero-Persistence:** IDs masked in RAM	Standard cloud storage (PII saved on disk)	Zero privacy (Photocopies, raw WhatsApp)
Indian Language Ingress	**10 Indian Languages** with script & voice audio	Static English & Hindi website forms	Verbal local dialect only (No digital tool)
Time to Final Activation	**3.8 Days** (Digital dispatch & automated provisioning)	14 to 21 Days (Paper manual verification)	5 to 7 Days (Unmanaged setup)
Customer Status Visibility	**Live 6-Stage Web/WhatsApp Tracker + Tech OTP**	Manual chase calls to 1800 call center	Unreachable technician mobile numbers
Channel Partner Model	**15% Recurring Commission** via dynamic QR	Direct sales force (Cuts out local dealers)	Fragmented resellers (No technology)

Why Jio Cannot Copy Quickly

Jio's enterprise architecture relies on rigid SAP GST pipelines and corporate direct sales teams. Discarding GST requirements demands rebuilding their DoT compliance stack from scratch.

Coexistence Over Cannibalization

Instead of fighting local cable operators, Accel pays them 15% recurring commissions via dynamic QR codes, converting former competitors into our local installation & maintenance franchise.

Telecom Physical Infrastructure Moat

Standalone SaaS software (Khatabook / Zoho) cannot provide physical last-mile fiber connectivity or cellular auto-failover. Accel owns the foundational physical infrastructure.

Quantified Business Impact & 12-Month Rollout

Before vs. After sales transformation and 12-month national scale roadmap

Funnel Economics Transformation

+38.8%
Quote Acceptance Rate
Jumps from 43.6% to 82.4% by eliminating the 5-day quote wait.

4.25x
Non-GST Conversion Lift
Conversion rises from 2.0% to 8.5% via instant Udyam lookup.

3.8 Days
Lead to Activation Cycle
Compressed by 78.4% from baseline 17.6 days via digital KYC.

₹3,250
Acquisition Cost (CAC)
Slashed by 78.3% from ₹15,000 via self-serve mobile flow.

LightGBM Lead Rescue Engine (Trained on 5,000 Leads)

AUC 0.84 MODEL

Retargets the **56.4% dropped quote cohort** within 15 minutes via enterprise WhatsApp API with 1-click pre-hydrated carts. Recovers **22.4% of abandoned prospects** (249 merchants from 5,000 leads), unlocking **₹48 Cr in recovered recurring ARR** at national scale.

Direct Labor & Operational Efficiency Gains:

Field Sales Rep Time / Customer	**24.0 Hours → 2.1 Hours (-91.2%)**
Customer Status Chase Calls	**3.2 Calls → 0.4 Calls (-87.5%)**
6-Month Customer Churn Rate	**14.2% → 6.8% (-52.1% via SLAs)**
Catalogue Rule Violation / Mis-Sell Rate	**15.2% (56/368) → 0.0% (-100% via CPQ DAG)**
Gross Margin on Bundles	**42.5% → 61.8% (+19.3% automated margin)**
Partner Dealer Commission	**15% Recurring via Instant QR Attribution**
Quote Generation Speed	**3 to 5 Days → < 60 Seconds (-99.9%)**
Payback Period on Sales CAC	**4.8 Months (vs 14.2 Months baseline)**

Bottom-Line Result: Direct sales acquisition cost slashed from ₹15,000 to ₹3,250, compressing CAC payback from 14.2 months to 4.8 months while freeing 22 hours of sales rep labor per closed account.

12-Month National Scale Strategy

Phase 1: Metros & Kirana Density (Months 1–3)

50K UNITS · ₹12 CR ARR

- Target 8 Tier-1 cities (Delhi NCR, Mumbai, Bengaluru, Hyderabad).
- Deploy Udyam Vyapar & Clinic packs via 15,000 retail store QR codes.
- Activate 50,000 merchants with instant paperless Udyam onboarding.
- Train 1,200 local cable operators as authorized 15% commission partners.

Phase 2: Industrial Clusters & Clinics (Months 4–6)

120K UNITS · ₹38 CR ARR

- Expand into 40 manufacturing towns (Surat, Coimbatore, Ludhiana, Peenya).
- Focus on Leased Line connectivity and multi-store Cloud CCTV packs.
- Capture 120,000 businesses with mission-critical 99.95% SLAs.
- Roll out automated field technician dispatch with OTP tracking across hubs.

Phase 3: All-India Bharat Scale (Months 7–12)

500K UNITS · ₹1,200 CR ARR

- Roll out across 500+ Tier-3/4 districts supported by 10 Indian languages.
- Integrate full voice biometrics and regional dialect speech synthesis.
- 12-Month Goal: **500,000 active small businesses** generating **₹1,200 Cr annualized ARR** at ₹3,250 blended CAC.

Strategic Airtel Ecosystem Synergies:

Airtel Payments Bank: Integrated QR soundbox for Kirana checkouts; daily merchant cash settlements automatically offset monthly Accel broadband subscriptions.

Airtel IQ WhatsApp API: Sub-second transactional quote push and LightGBM automated retargeting with 1-click pre-hydrated carts. Recovers 22.4% of abandoned prospects.

Nxtra Data Centers: Sovereign Tier-4 Indian data facilities guarantee 100% DPDP Act 2023 compliance with zero offshore data transit or foreign jurisdiction exposure.

Airtel 5G Plus QoS Slicing: Dedicated network QoS packet queues ensure zero latency jitter during evening peak payment hours (7–9 PM), preventing lost retail checkouts.

Airtel Merchant Finance: Pre-approved working capital micro-loans up to ₹5 Lakhs for active Accel businesses based on automated broadband uptime and UPI transaction history.

Executive Verdict: **Project Udyam Digital transforms Accel Business from a slow, paper-bound corporate carrier into India's default digital operating system for small enterprises—slashing CAC by 78%, lifting non-GST conversions 4.25x, and unlocking ₹1,200 Cr ARR in 12 months.**